tips = df

Seaborn

→ !pip install seaborn or add to req.txt

```
>>> import seaborn as sns
>>> sns.load_dataset('tips')
```

loading a dataset.

① Creating scatter...

import matplotlib.pyplot as plt.

```
>>> sns.scatterplot(x='C1', y='C2', data=tips)
>>> plt.title('%0 %0')
```

column name1 — column name 2 — entire data

Line Plot

```
>>> sns.lineplot(x='C1', y='C2', data=tips)
>>> plt.title('Line Plot')
>>> plt.show()
```

Note: reset_index() removes the category of grouping from the result

Seaborn lineplot

→ treats y as obs, optionally grouped by x

→ Calculates mean & confidence interval

→ Draws shaded area for CI.

tips = df

/ Categorical → Bar Plots

```
>>> sns.barplot(x='col1', y='col2', data=tips)
    plt.title(..)
    plt.show()
```

→ Box plots

```
>>> sns.boxplot(x='c1', y='c2', data=tips)
```

→ Violin Plot

```
>>> sns.violinplot(x='c1', y='c2', data=tips)
```

For numerical → Histograms

To check the distribution of data.

```
>>> sns.histplot(tips['column_name'], bins=10, kde=True)
```

no of. bins

data column to be used

distribution curve display

$\boxed{\text{tips} = df}$

→ Same KDE curve,

```
>>> sns.kdeplot(tips['total_bill'], shade=True)
```

→ Gives distribution curve, shaded.

Pair Plot

```
sns.pairplot(tips)
```

→ considers only numerical values.

→ gives correlation between different values.

Heat Map → (for correlation)

⇒ (use numerical features)

```
corr = tips[['total_bill', 'tip', 'size']].corr()
```

finding the correlation for

	tb	tip	size
tb	-	-	-
tip	-	-	-
size	-	-	-

```
sns.heatmap(corr, annot=True, cmap='coolwarm')
```

heatmap

annotations = True

colourmap

df = Sales - data

Q → Total sales by product.

```
plt.figure(fig size = (6,6))
sns.barplot(x='Prod', y='T_R', data=df,
            estimator=sum)
plt.title()
   xlabel
plt.show()
```

put any agg()
eg: sum, mean, ...

~~alternate → group data, reset index,~~
~~→ reset index~~

alternate
→ group data on Prod
→ find sum of TR
→ reset_index / make a df
→ plt.bar(x, y, color ...)
x: categories, y: values